Piecewise linear graphs

1

a $a = -1$

b -2

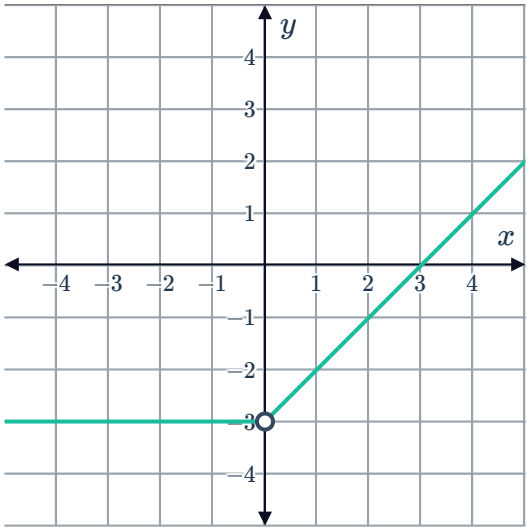
2

a
$$y = \begin{cases} \frac{3}{2}x + 3 & \text{if } x \leq 0 \\ 3 - x & \text{if } x < 0 \leq 3 \\ 2x - 6 & \text{if } 3 < x < 5 \end{cases}$$

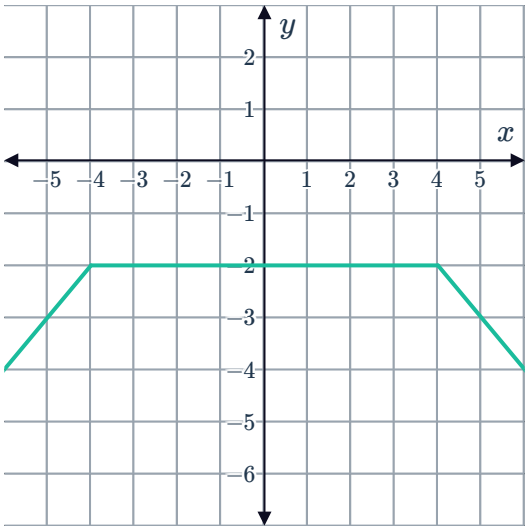
b
$$y = \begin{cases} x + 1 & \text{if } -5 \leq x \leq 1 \\ 1 - x & \text{if } 1 < x \leq 4 \end{cases}$$

3

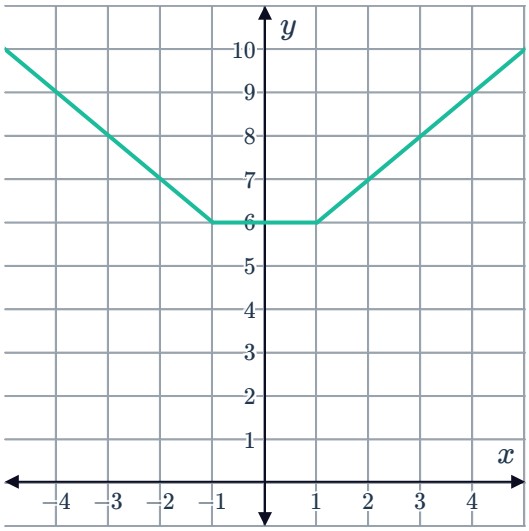
a



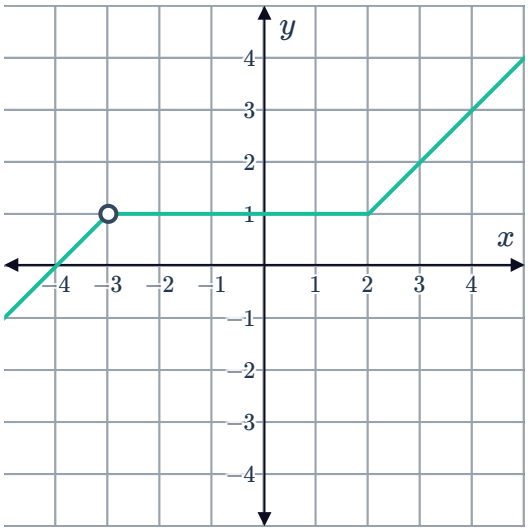
b



c

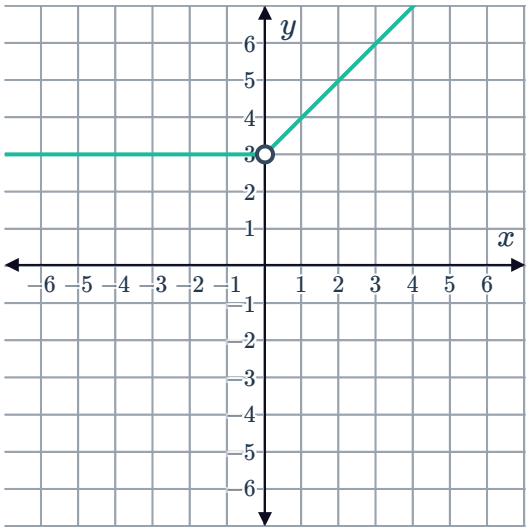


d

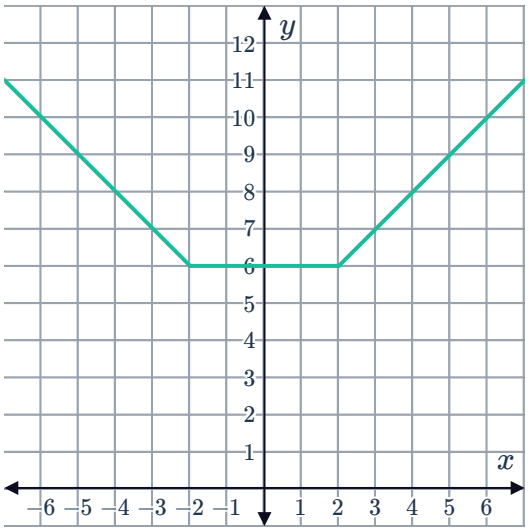


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

e



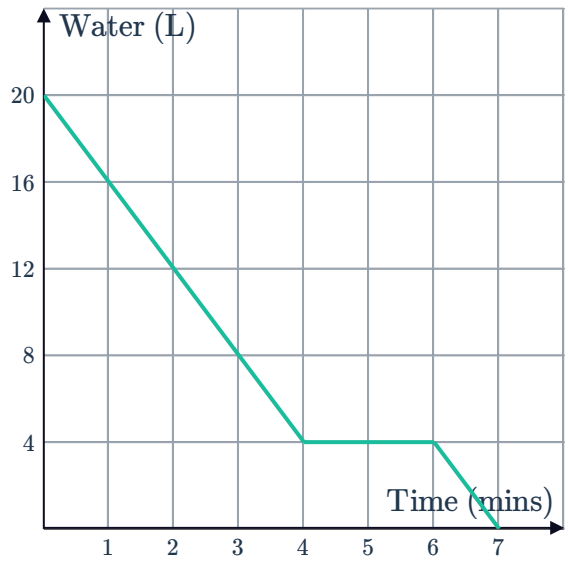
f



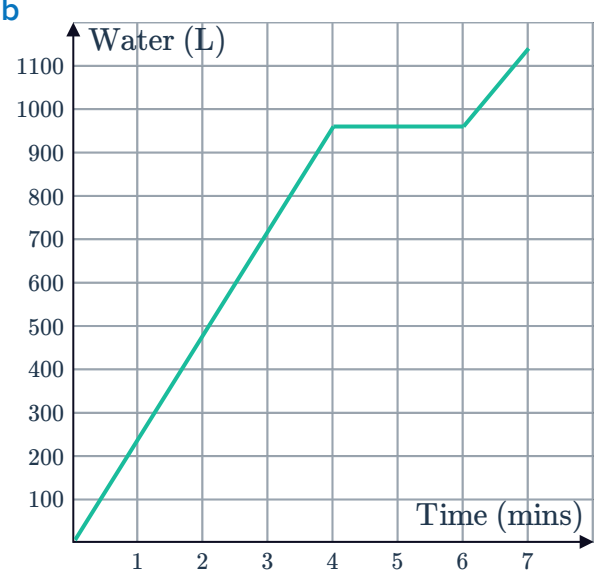
Piecewise Models

4

a



b



5

a 16 L

b The driver filled the tank.

c 16 L

d 12 pm

6

a 2 km

b 1 pm

c 4 km

7

a 2.5 km

b 3:00 pm

c 3.5 km

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

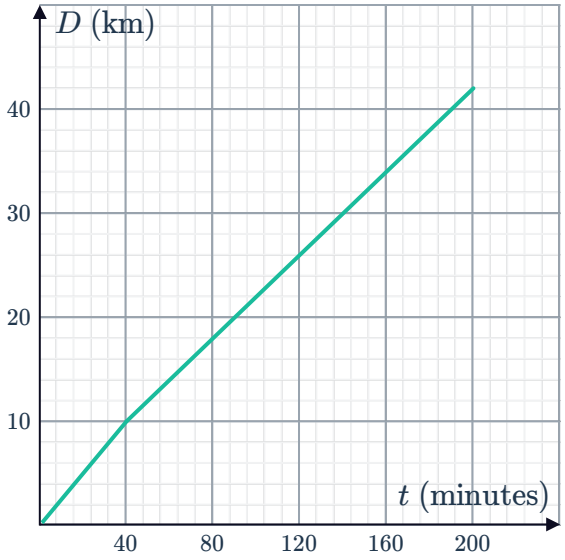
a 200 L **b** $V = \frac{40}{9}t$ **c** $t = 95$

9

a $D = \begin{cases} 0.25t & \text{when } 0 \leq t < 40 \\ 2 + 0.2t & \text{when } 40 \leq t \leq 200 \end{cases}$

b **i** 5 km **ii** 10 km **iii** 14 km **iv** 42 km

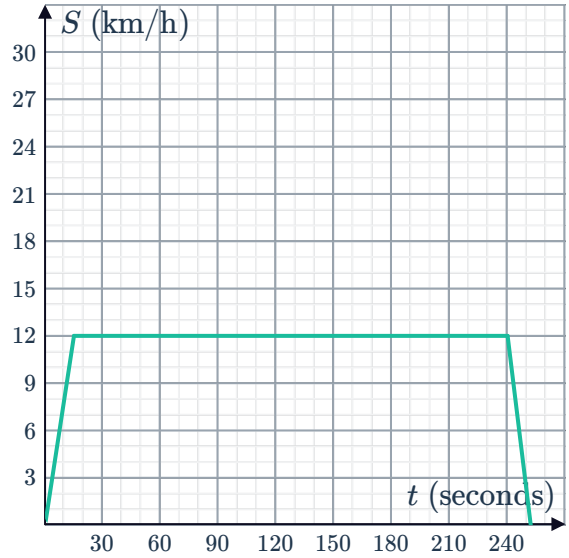
c



10

a **i** 8 km/h **ii** 12 km/h **iii** 12 km/h **iv** 2 km/h

b



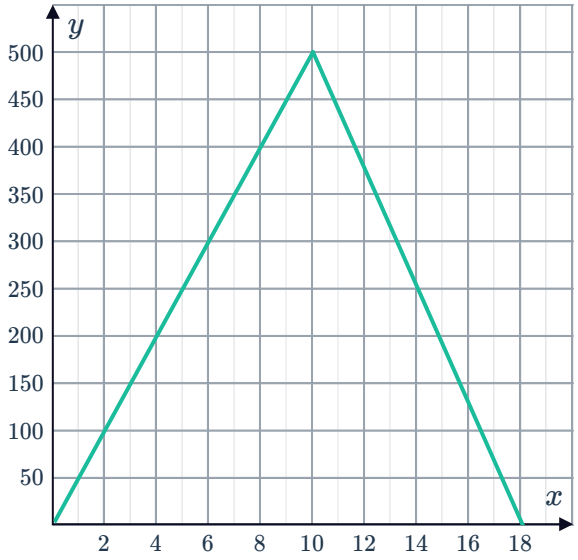
11

a 12 **b** $E = 14$ **c** $24 < t \leq 36$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12

a



b 748 m

Step graphs

13

a $y = 1$

b $y = -3$

14

a $y = -5$

b $y = 8$

15

a $y = 3$

b $y = -4$

Step graph models

16

a

i \$1

ii \$5.50

b 175 g

17

a \$10

b 4.75 hours

c \$6

18

a \$9

b \$12

c 5 hours

19

a 7 pm

b No

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20

a 1 hour b $5 c $6

21

a $2400 b $2000 c 1 hour

22

a $3 b $3 c 15

23

a $1.70 b $1.30 c 4 minutes d $0.20

e $0.70

24

a

i $10 ii $12 iii $6 iv $14

b Less than 2 kg

c

i Less than 1 kg ii Less than 1 kg

iii Less than 0.5 kg iv Less than 4 kg

25

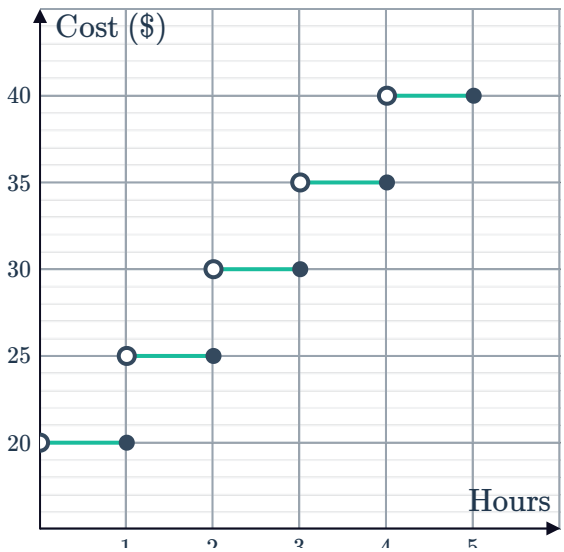
a $2.50 b $8 c $1.50 d 16 km

e $55

26

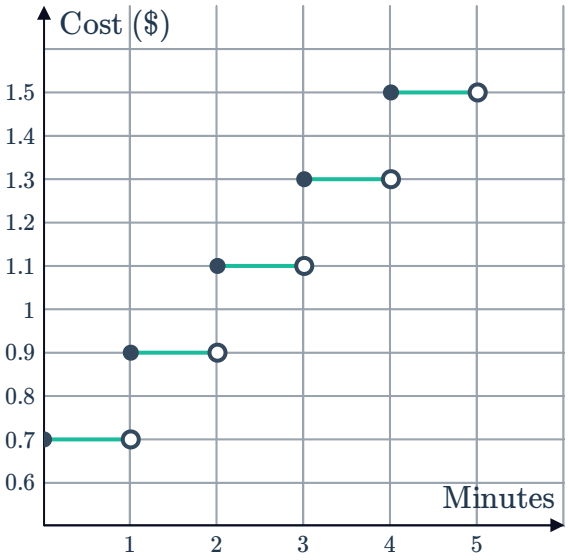
a $2.10 b $6 c $51.30

27

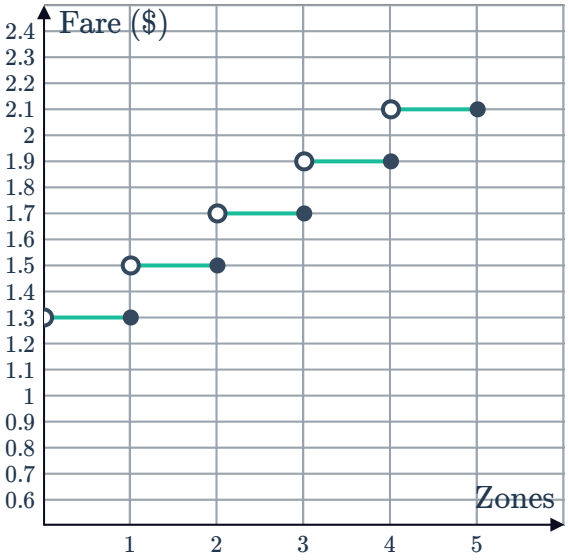


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

28



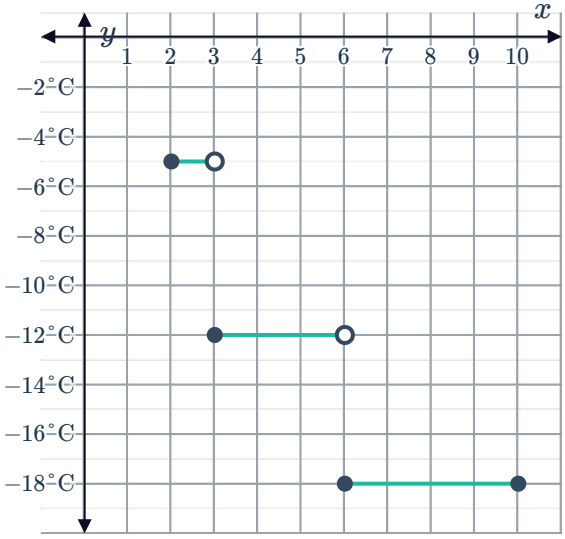
29



30

a $$y = \begin{cases} -5^\circ\text{C}, & 2 \leq x < 3 \\ -12^\circ\text{C}, & 3 \leq x < 6 \\ -18^\circ\text{C}, & 6 \leq x \leq 10 \end{cases}$$

b

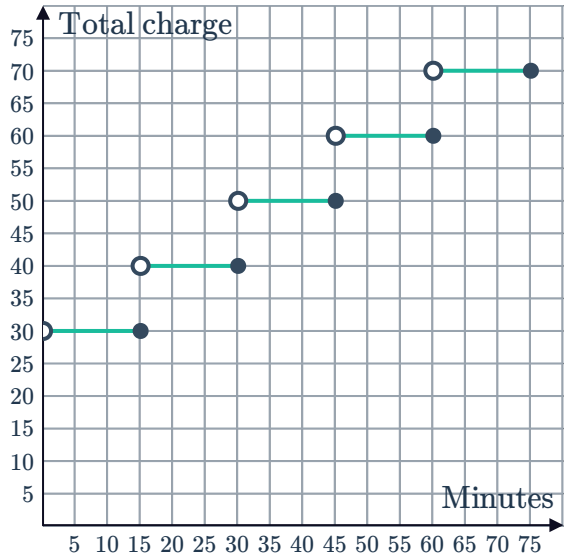


c -12°C

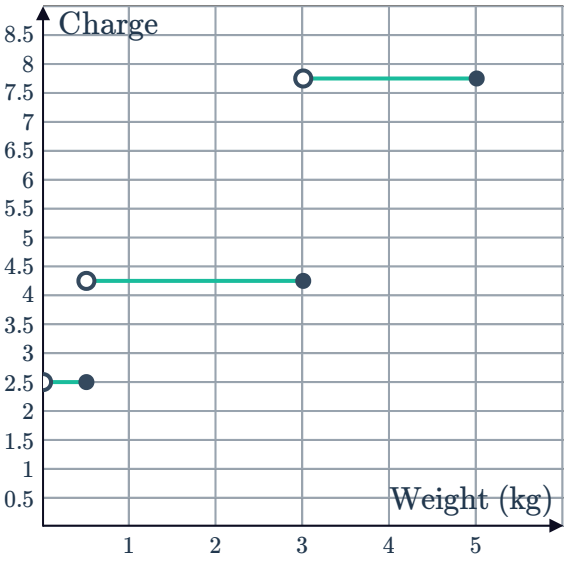
d 3 pm

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31



32



33

a $1200 **b** $48 999 **c** $1200

© ClipSAT: By Mr. Mohamed Abdallah +966597688647